

MACHINE LEARNING PROJECT

Bank Marketing Campaign Deposit Subscription

*Predicting whether a customer will subscribe to a term deposit using
Machine Learning models*

Team

THE IDEA

Why This Project?

Banks run costly outbound campaigns to convince clients to open a term deposit. Calling every customer wastes time and money — most people say no.

Our goal: build a Machine Learning model that predicts, which customers are most likely to subscribe — so the bank can focus its effort where it counts.

THE DATASET

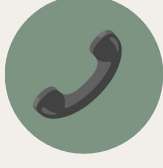
Features We Worked With

bank.csv — direct marketing data from a Portuguese bank, target column: deposit (yes/no)



Client Profile

- age
- job
- marital
- education
- default (credit)
- balance
- housing loan
- personal loan



Current Campaign

- contact type
- day
- month
- duration
- campaign (contacts made)

THE TEAM

Who Did What

F

Fatma

Exploratory Data Analysis (EDA) & Data Cleaning

R

Rehab

Logistic Regression & K-Nearest Neighbors (KNN)

H

Hagar

Decision Tree & Random Forest Models

M

Marwan

Boosting Models (XGBoost, LightGBM)

Key Findings from Exploration

1

Hidden Missing Values

job, education, contact and poutcome used the label "unknown" instead of real NaNs — treated as a category.

2

Th

-1 meant "r
was_contact

3

Balanced Target

deposit is close to a 50/50 split between yes and no, so accuracy alone is a fair-but-not-sufficient metric.

4

Jo

students an
entreprene

PREPARING THE DATA

Preprocessing & Feature Engineering

1

Fix pdays

Created
was_contacted_befo
re flag; replaced
sentinel -1 with 0.

2

Cap Outliers

Capped campaign at
the 99th percentile
instead of deleting
rows.

3

Encode

Binary yes/no
columns mapped to
1/0; nominal
columns one-hot
encoded.

MODELING

Six Algorithms, One Goal

Every model was evaluated on Accuracy, Precision, Recall, F1-score and ROC-AUC, with Recall prior extra call.

1

Logistic Regression

Linear baseline; tested default vs. class-balanced weighting.

2

K-Nearest Neighbors

Tuned K, distance metric and weighting via validation.

4

Random Forest

Ensemble of trees; tuned via RandomizedSearchCV for Recall.

5

XGBoost

Gradient boosting with scale_pos_weight for imbalance.

RESULTS

Final Model Comparison

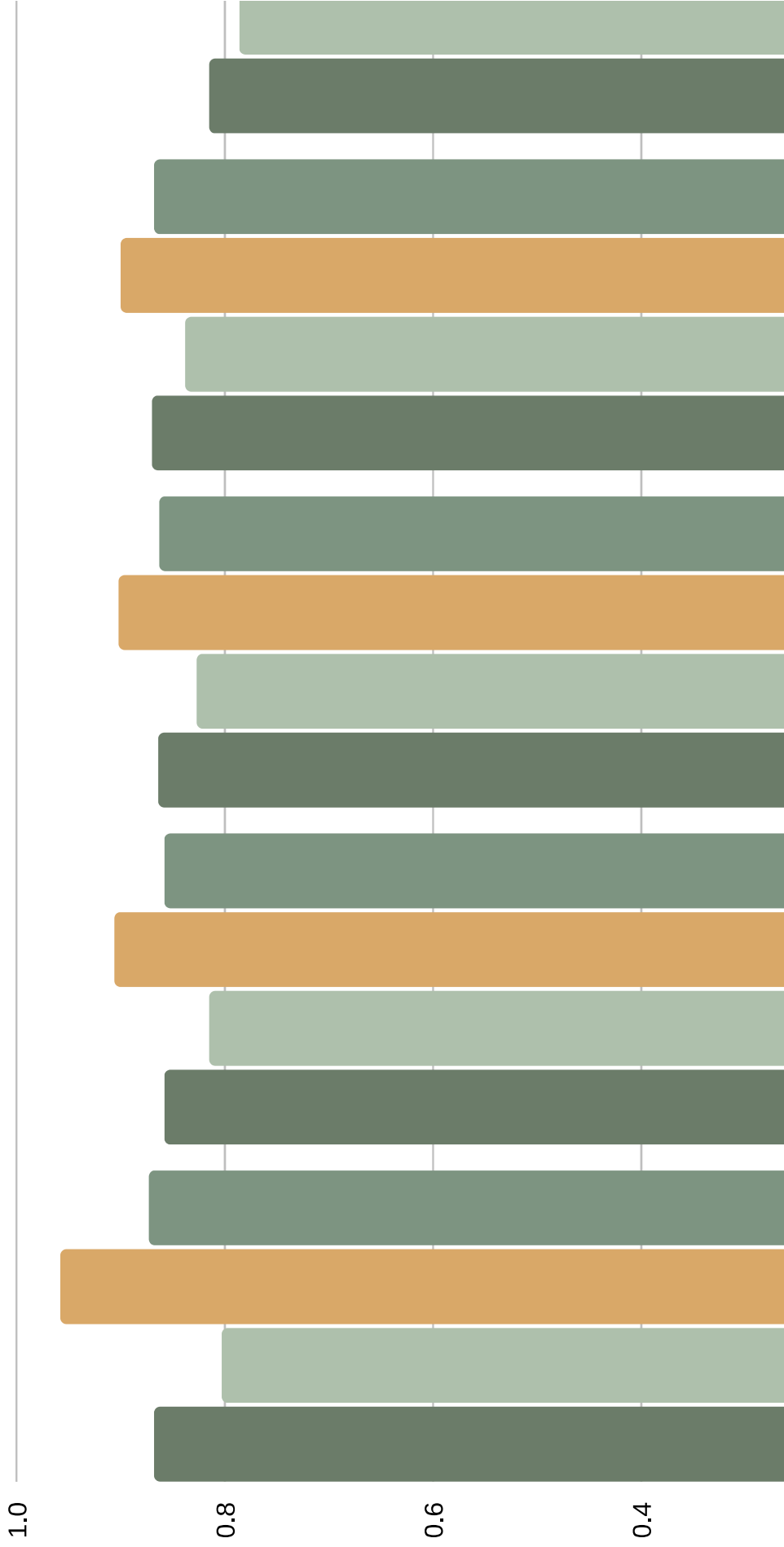
Sorted by Recall — our primary metric

Model	Accuracy	Precision
RF + LightGBM Ensemble	0.868	0.803
Random Forest (Recall-Opt.)	0.858	0.815
LightGBM (Recall-Opt.)	0.864	0.827
XGBoost (Recall-Opt.)	0.870	0.838
Decision Tree (Final)	0.815	0.786
Logistic Regression (Balanced)	0.829	0.821

RESULTS

Model Performance at a Glance

● Accuracy ● Precision ● Recall



THE WINNER

RF + LightGBM Ensemble

Soft-voting ensemble of the two strongest tree-based models, threshold-tuned for accuracy.

95.8%

Recall

86.8%

Accuracy

87.3%

F1-score

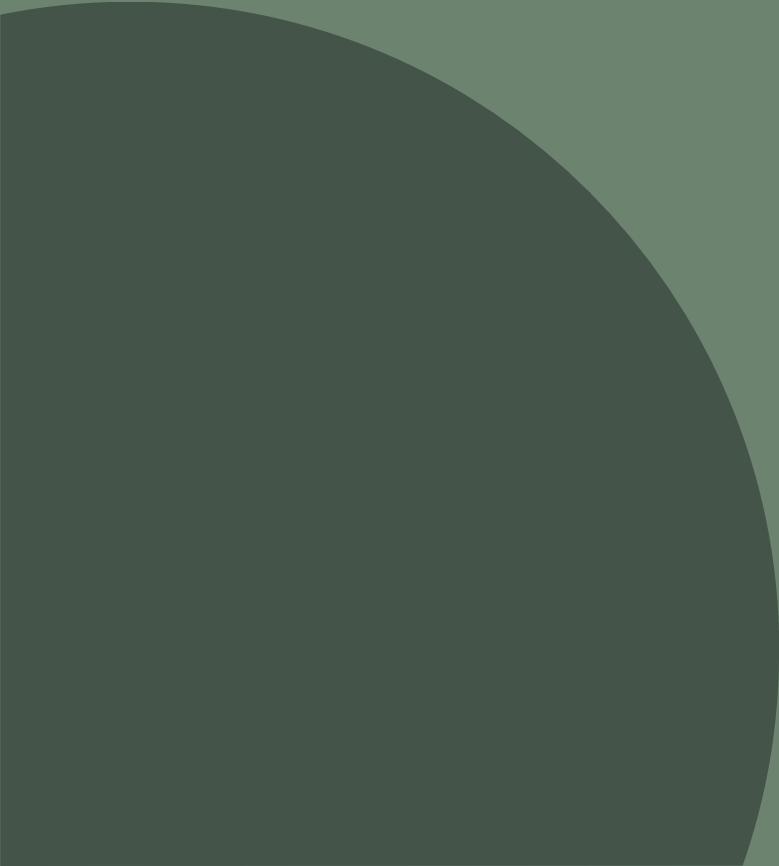
Why Recall came first

Missing a real subscriber costs a lost sale; an extra unnecessary call costs a few

Pipeline

Model Pipeline

- 1 Load & clean bank.csv
- 2 Engineer was_contacted_before, cap outliers
- 3 Train/test split (80/20, stratified)
- 4 One-hot encode + StandardScaler (fit on train only)
- 5 Train & tune 6 models with GridSearch / RandomizedSearch
- 6 Score on Accuracy, Precision, Recall, F1, ROC-AUC



Thank You

Bank Marketing Campaign

OUR TEAM

- Fatma Khalil
- Hassan Arafat